

3969. Valid Subarrays With Matching Sum Digits 1

input: num = [1, 100, 1]

Sub-arrays	x =	Sum	1st digit	last digit
nums[0]	1	1	1	1 count 1
nums[0, 1]	1, 100	101	1	1 count 2
nums[0, 1, 2]	1, 100, 1	102	1	2
nums[1]	100	100	1	0
nums[1, 2]	100, 1	101	1	1 count 3
nums[2]	1	1	1	1 count 4

Count where 1st & last digits are same

output = 4.

for getting sub-arrays

```
int count = 0;
```

```
for (int start = 0; start < nums.length; start++) {  
    int sum = 0;  
    for (int end = start; end < nums.length; end++) {  
        sum = sum + nums[end];  
    }  
}
```


for taking first digit of sum

```
int firstDigit = sum;
while (firstDigit >= 10) {
    firstDigit = firstDigit / 10;
}
```

for taking last digit of sum

```
int lastDigit = sum % 10;
```

for check 1st & last digit is equal to x?

```
if (firstDigit == x && lastDigit == x) {
    count ++;
```

```
}
```

```
}
```

```
}
```

```
return count;
```


Dry Run -

input : nums [0, 100, 1], x = 1

output = 4

int count = 0;

reverse
var
④ 3 4 3
③ 2 4 3
② 1 4 3
① 0 4 3

for (int i = 0; i < nums.length; i++) {

int sum = 0;

③① j = 2

for (int j = i; j < nums.length; j++) {

①② 101 + nums[2]

①② 1 + nums[1]

①① 0 + nums[0]

sum = sum + nums[j]; sum = 102

②③ 100 + nums[1]
sum = 100 + 1
sum = 101

③① 0 + nums[2]
sum = 1

②② F = 101

①② F = 101

①① F = 1

①③ F = 102

②① F = 100

③① = F = 1

while (firstDigit >= 10) {

②② 101 >= 10
101 / 10 = 10
F = 10

③① 100 >= 10
100 / 10 = 10
F = 10

③② 102 >= 10
102 / 10 = 10
F = 10

③③ 101 >= 10
101 / 10 = 10
F = 10

③④ 10 >= 10
10 / 10 = 1
F = 1

firstDigit = firstDigit / 10;

101 / 10 = 10
F = 10

102 / 10 = 10
F = 10

10 / 10 = 1
F = 1

③① 100 / 10 = 10
F = 10

③② 10 / 10 = 1
F = 1

③③ 10 / 10 = 1
F = 1

③④ 10 / 10 = 1
F = 1

$$\begin{aligned} (3)(1) \quad 107/10 &= 10 \quad L=1 \\ (2)(1) \quad 101/10 &= 10 \quad L=1 \\ (2)(1) \quad 100/10 &= 10 \quad L=0 \end{aligned}$$

$$\begin{aligned} (1)(3) \quad 102/10 &= 10 \quad L=2 \\ (1)(2) \quad 101/10 &= 10 \quad L=1 \end{aligned}$$

int lastDigit = sum % 10; 01 % 10 L=1

if (firstDigit == x && lastDigit == x) {

$$(2)(1) \quad 1 == 1 \quad \&\&$$

$$(2)(2) \quad 1 == 1 \quad \&\&$$

$$(3)(1) \quad 1 == 1 \quad \&\&$$

count ++; count = 1 2 3 4

count = 4

}
}
}

return count; return 4

}
}